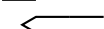
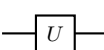
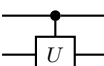
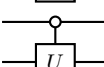
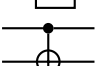
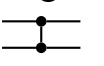
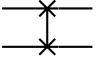
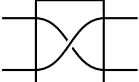
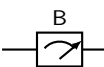
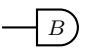


7. Digital quantum circuits (pictorial)

7A. Basic elements

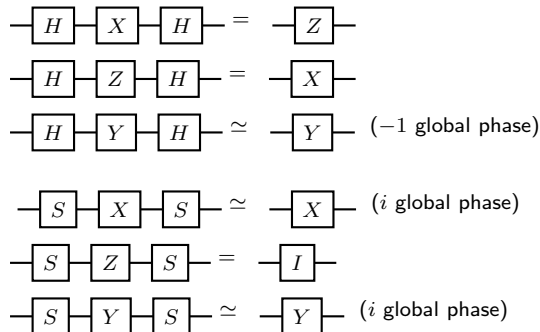
Quantum wire	
Quantum wire bundle (n qubits)	
Quantum wire bundle (alternate)	
Classical wire	
Entangled bit (ebit; Bell pair)	
Quantum gate $\hat{U}$	
Control gate U (control on $ 1\rangle$)	
Control gate U (control on $ 0\rangle$)	
Control-X (cNOT)	
Control-Z (cZ) ¹	
Swap gate	
Swap gate (alternate)	
Measurement in basis B	
Measurement in basis B (alt)	

¹controlled-Z operation is "symmetric" in the roles of control and target; hence the circuit representation by two dots.

7B. Circuit identities

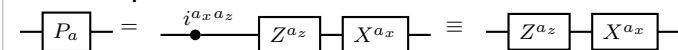
Proofs and checks: See (C69) and Mathematica/2021 RC Learn experiment/2022-11.1 circuit identities basic.nb

Pauli operator basis change

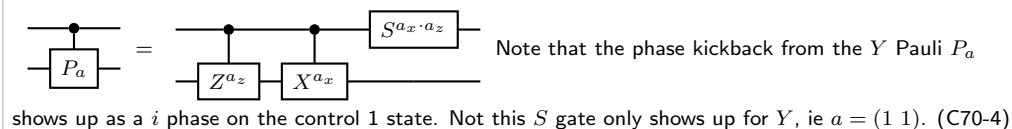


Basic and super useful

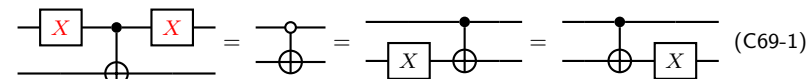
Pauli decomposition



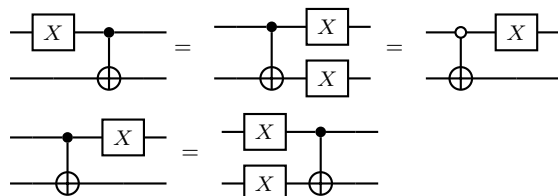
Pauli decomposition $P_a = i^{a_x a_z} X^{a_x} Z^{a_z}$ with $a = (a_x, a_z)$, see Sec. 3C. Note that for gates $P_a \cdot P_a^\dagger$, the $i^{a_x a_z}$ drops out. The global phase $i^{a_x a_z}$ for non control Pauli P_a can be ignored. It only applied to $a = (1, 1)$ for $P_a = Y$ (C70-4)



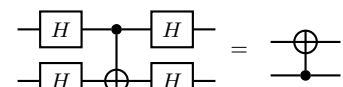
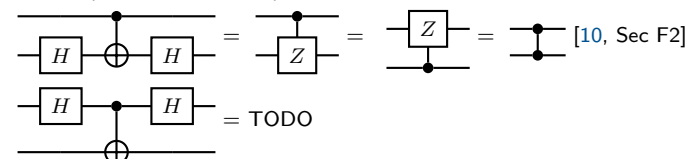
cX + X



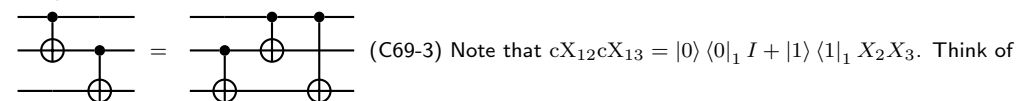
"X travels forwards" from control to target (C69-2)



cX + H (Control-Z cZ gate)



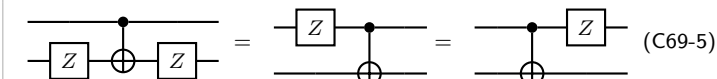
Multiple cNOTs



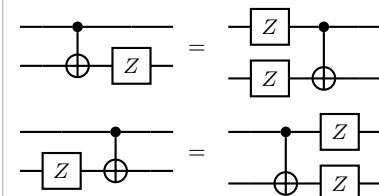
having to push through the X on the control-2 only when control-1 is $|1\rangle\langle 1|$, then pushing X through cX_{23} yields X travels forwards and gives an X on each of the 2-3 wires.

Controlled-NOT Gate (cNOT, cX)

cX + Z



"Z travels backwards" from target to control (C69-5 / see above)



cX + S (Control-Y cY gate)

